Builder Pattern

Constructing complex objects step by step with precision.

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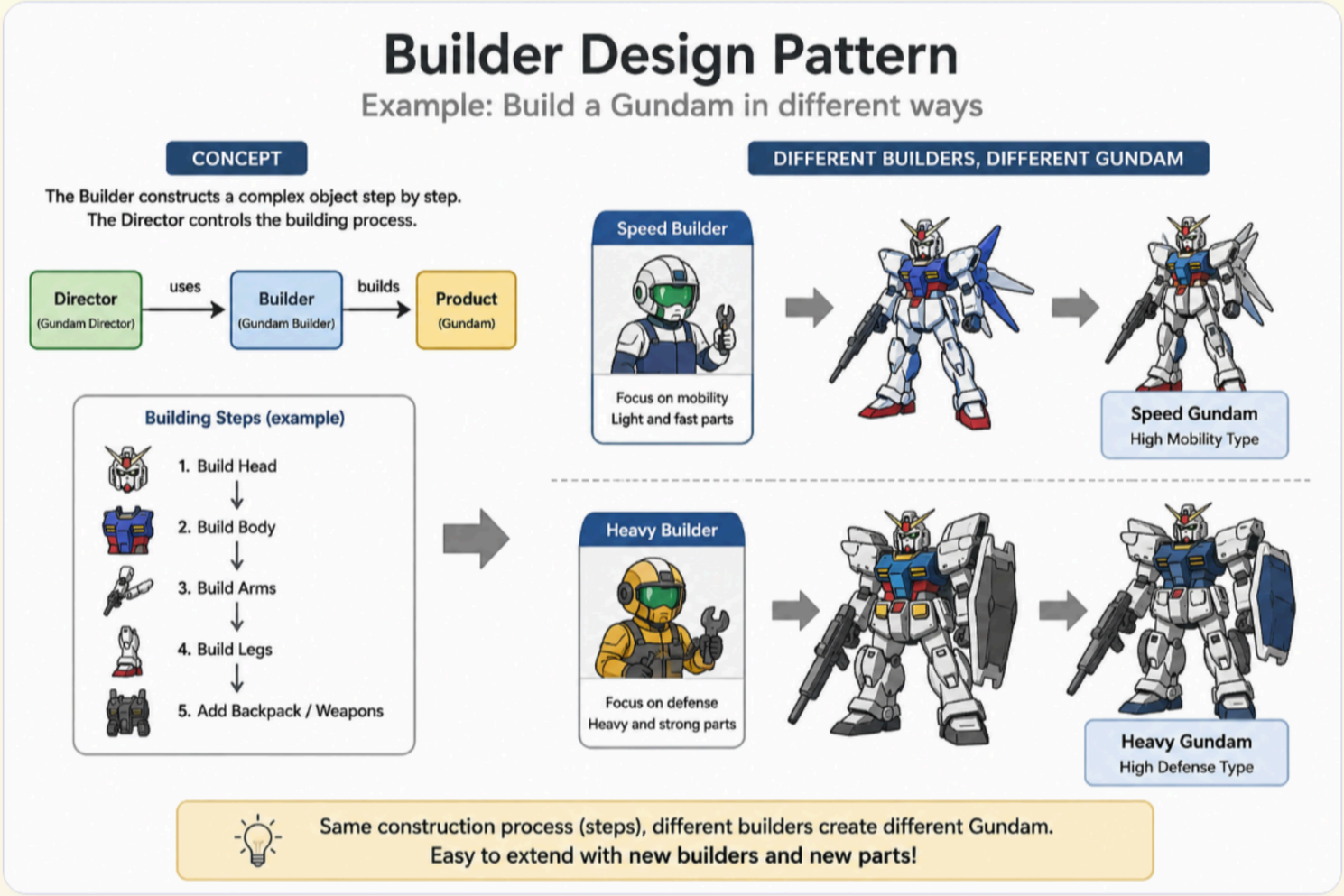
What is the Builder Pattern?

Builder is a creational design pattern that lets you construct complex objects step by step.

It extracts the object construction code out of its own class and moves it to separate objects called *builders*. This pattern allows you to produce different types and representations of an object using the same construction code.

The Builder Concept



The Telescoping Constructor Problem

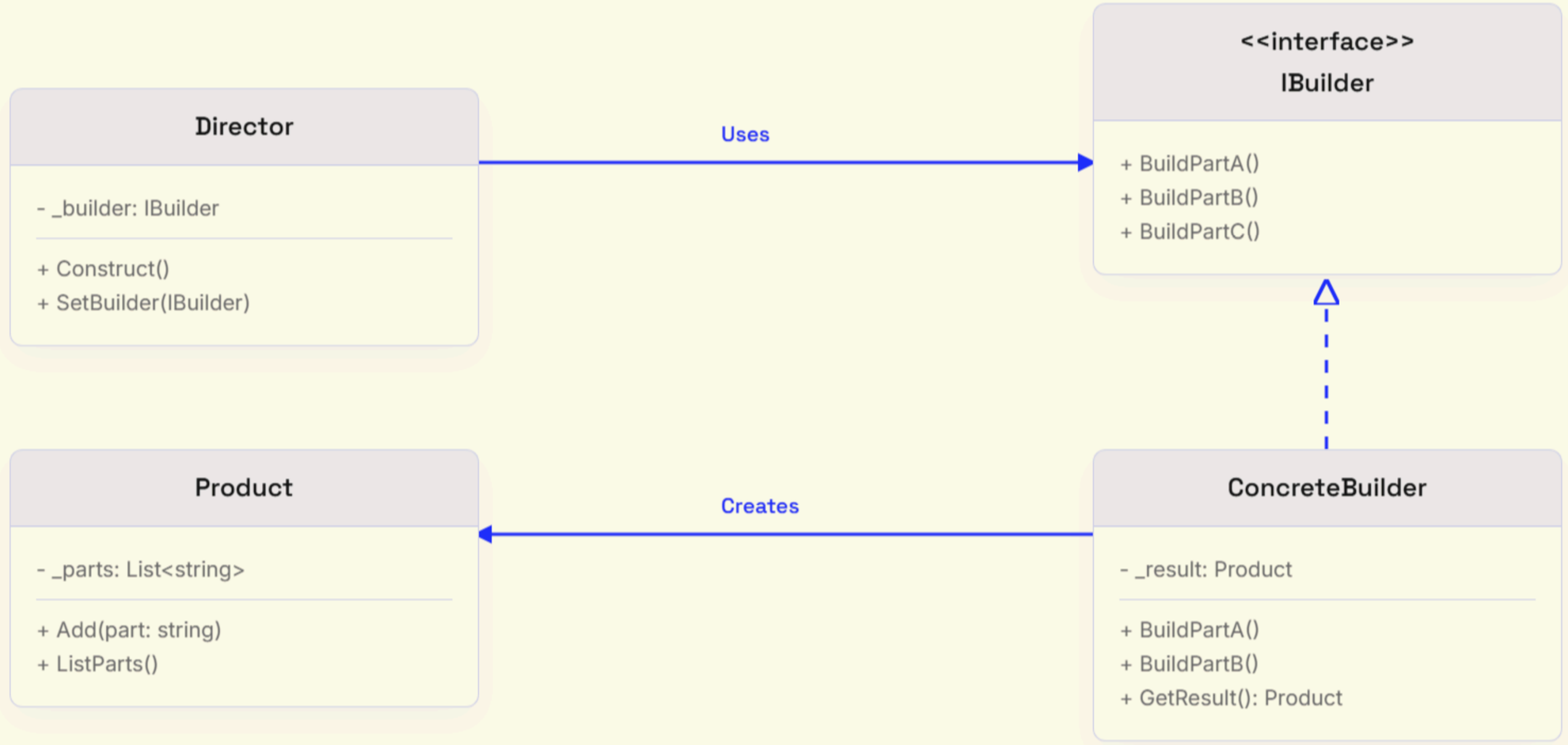
The Problem

- Massive constructors with dozens of parameters.
- Most parameters are unused, leading to ugly code like `new House(4, 2, true, false, null)`.
- Extremely difficult to remember the order of parameters.
- Creating subclasses for every possible configuration causes an explosion of classes.

The Solution

- Extract construction logic into dedicated **Builder** classes.
- Construct objects dynamically using clean, fluent interfaces.
- Only execute the specific steps you actually need.
- Defer final instantiation until the entire configuration is ready.

C# Class Diagram



The Anti-Pattern

Creating a complex object requires specifying every possible detail in a massive constructor call, often passing many `null` or `false` values which makes the code brittle.

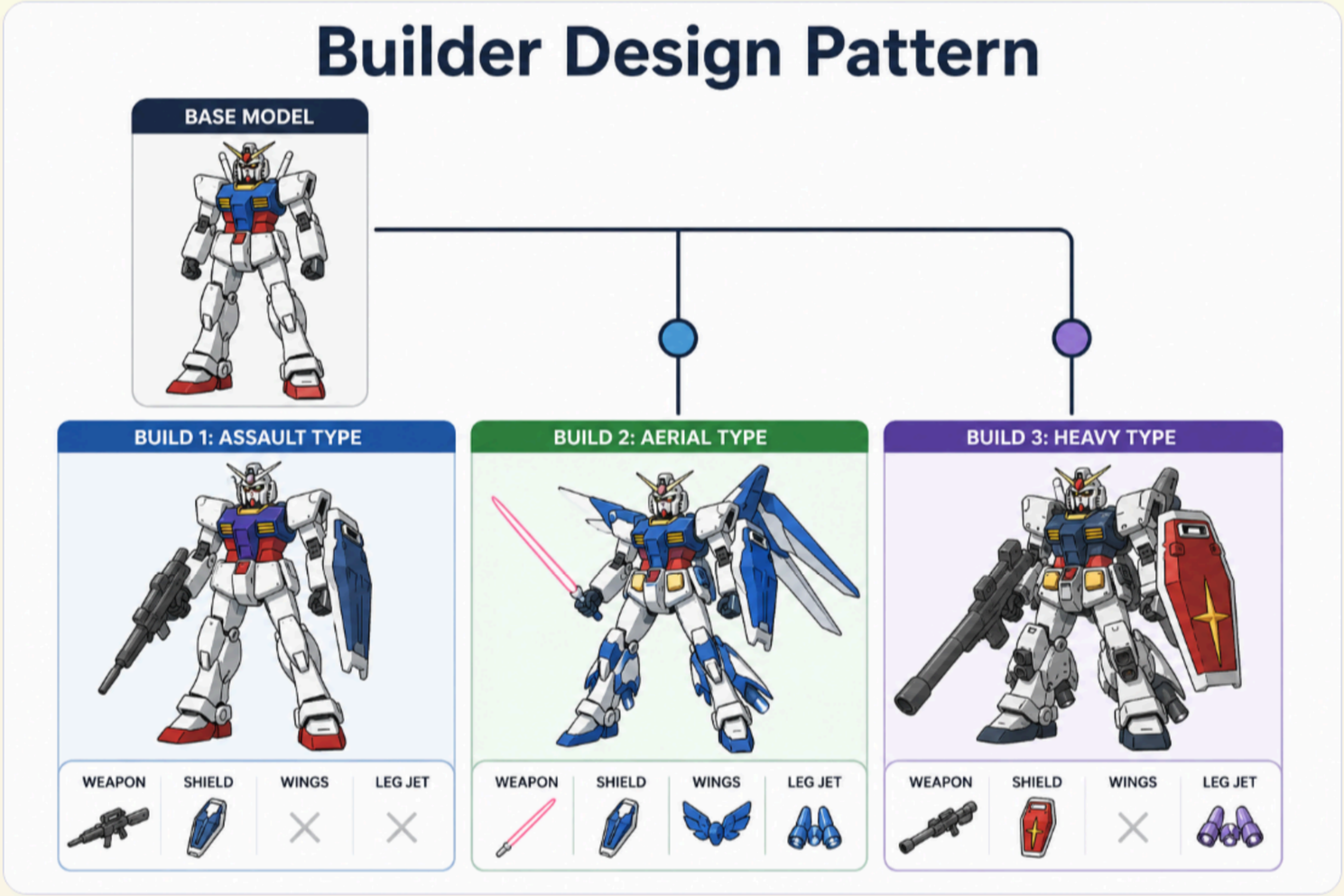
```
// Hard to read, easy to mess up parameter order, completely rigid.
Gundam heavyGundam = new Gundam(
    "Heavy Sensor Head",      // head
    "Heavy Armor Body",       // body
    "Heavy Arms",             // arms
    "Reinforced Legs",        // legs
    "Heavy Shield",           // shield
    "Bazooka",                // weapon
    null                      // hasBackpack
);
```

Fluent Builder Construction

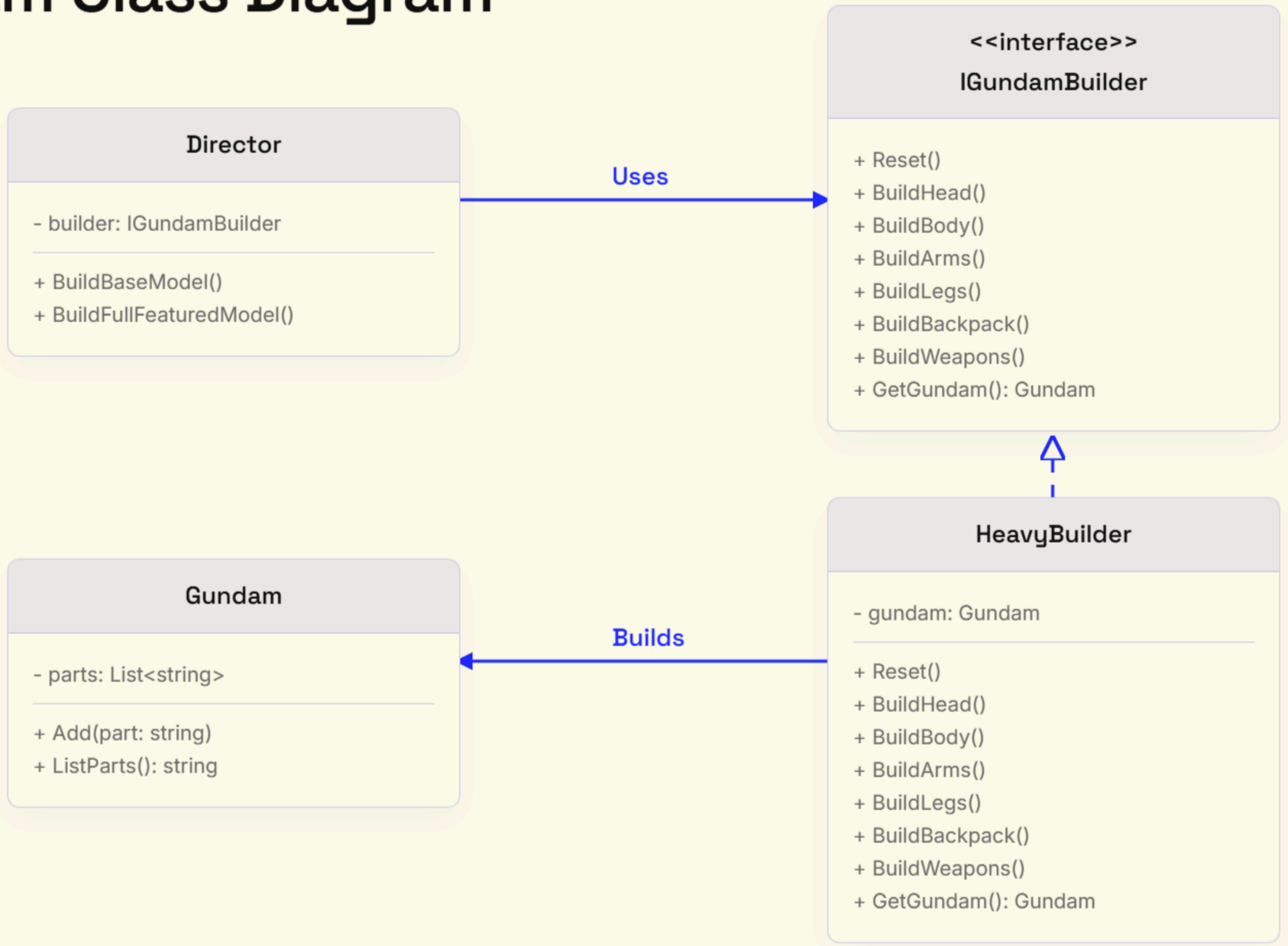
We use a Builder to construct the house step-by-step. The code is self-documenting and we only invoke the steps we actually need.

```
// Clean, readable, and intent-revealing fluent interface.
HeavyBuilder gundamBuilder = new HeavyBuilder();
Gundam heavyGundam = gundamBuilder
    .BuildHead()
    .BuildBody()
    .BuildArms()
    .BuildLegs()
    .BuildBackpack()
    .BuildWeapons()
    .Build();
```

Three Unique Builds



Gundam Class Diagram



Product

```
// The complex product we want to build
public class Gundam
{
    private readonly List<string> _parts = new();

    public void Add(string part)
    {
        _parts.Add(part);
    }

    public string ListParts()
    {
        return "Gundam Parts:\n- " + string.Join("\n- ", _parts);
    }
}
```

Interface

```
// The builder interface specifying methods for creating the different parts
public interface IGundamBuilder
{
    void Reset();
    void BuildHead();
    void BuildBody();
    void BuildArms();
    void BuildLegs();
    void BuildBackpack();
    void BuildWeapons();
    Gundam GetGundam();
}
```

Concrete Builder

```
// Concrete builders implement steps defined in the interface
public class HeavyBuilder : IGundamBuilder
{
    private Gundam _gundam = new();

    public void Reset() => _gundam = new Gundam();
    public void BuildHead() => _gundam.Add("Heavy Sensor Head");
    public void BuildBody() => _gundam.Add("Heavy Armor Body");
    public void BuildArms() => _gundam.Add("Heavy Arms");
    public void BuildLegs() => _gundam.Add("Reinforced Legs");
    public void BuildBackpack() => _gundam.Add("Heavy Shield");
    public void BuildWeapons() => _gundam.Add("Bazooka");

    public Gundam GetGundam()
    {
        var result = _gundam;
        Reset(); // Builders should be ready to start producing another product
        return result;
    }
}
```

Director

```
// The Director controls the order of the building steps
public class Director
{
    private IGundamBuilder _builder;
    public IGundamBuilder Builder { set => _builder = value; }

    public void BuildBaseModel()
    {
        _builder.BuildHead();
        _builder.BuildBody();
        _builder.BuildArms();
        _builder.BuildLegs();
    }

    public void BuildFullFeaturedModel()
    {
        BuildBaseModel();
        _builder.BuildBackpack();
        _builder.BuildWeapons();
    }
}
```

Usage

```
// Client Code Usage
var director = new Director();
var heavyBuilder = new HeavyBuilder();
director.Builder = heavyBuilder;

director.BuildFullFeaturedModel();
var heavyGundam = heavyBuilder.GetGundam();
Console.WriteLine(heavyGundam.ListParts());
```

Pros & Cons

Pros

- **Step-by-step:** Construct objects iteratively, defer construction steps, or run steps recursively.
- **Reusability:** Reuse the same construction code when building various representations of products.
- **Single Responsibility:** Isolate complex construction code from the business logic of the product.

Cons

- **Complexity:** The overall complexity of the code increases since the pattern requires creating multiple new classes (Builder, ConcreteBuilders, Director).
- **Dependency:** The client must be aware of the specific builder types if they aren't hidden behind a Director.

Thank You

Questions?